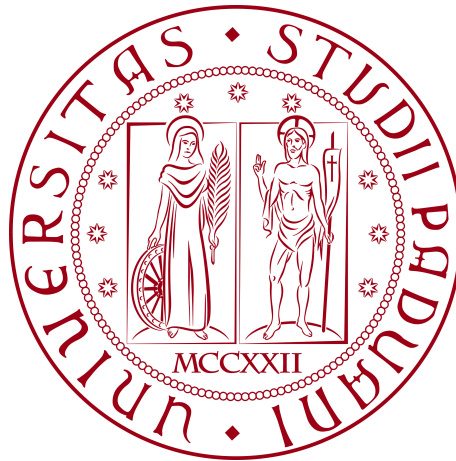


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When Does Structural Reasoning Help? Structural Generalization in Neurosymbolic Learning

Master's Thesis

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“We can only see a short distance ahead, but we can see plenty there that needs to be done.” — Alan Turing

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This thesis marks the conclusion of an important chapter of my academic journey and, at the same time, the culmination of a path that has allowed me to explore Artificial Intelligence in greater depth. During these years, I have had the opportunity to learn not only about algorithms, models, and technologies, but also about the broader questions surrounding the development and use of intelligent systems. The rapid evolution of Artificial Intelligence

makes it difficult to predict where this field will lead in the future, but it also makes clear that there is still much to discover, understand, and build.

As I reach the end of this Master's degree, I look back with gratitude at the people who supported me and at the experiences that shaped this journey. I hope to carry forward the knowledge, curiosity, and passion developed during these years, and to continue exploring the possibilities of a field that is still only beginning to reveal its full potential.

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Abstract

This thesis investigates the use of Structured Logic Tensor Networks (LTNs) for neuro-symbolic learning. The goal is to combine neural learning with explicit background logical knowledge. The proposed approach addresses an AI classification task with limited and heterogeneous training data. Entities and concepts are represented through learnable continuous embeddings. Logical relations and domain constraints are expressed using first-order logic. These logical formulas are converted into differentiable constraints using fuzzy logic semantics. The model jointly learns from labeled examples and the available background knowledge. A structured representation is used to organize concepts, relations, and logical dependencies. This allows domain knowledge to guide the learning process rather than relying only on observed data. The approach is evaluated against conventional neural network baselines. Experiments investigate both predictive accuracy and logical consistency of the learned model. Additional experiments analyze the effect of increasing or reducing the amount of background knowledge. The results examine whether logical constraints improve generalization, particularly with limited training data. The thesis also discusses computational and modeling challenges associated with differentiable logical reasoning. Overall, the study demonstrates the potential of Structured LTNs for integrating learning and reasoning in AI systems.

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Chapter 1

Introduction

Artificial intelligence systems are increasingly required not only to learn from data, but also to reason about the knowledge acquired from their environment. Modern machine learning methods have demonstrated remarkable capabilities in processing complex and high-dimensional data, including images, natural language, and other forms of unstructured information. However, learning directly from data does not by itself provide an explicit representation of knowledge or a principled mechanism for expressing logical constraints. Conversely, symbolic artificial intelligence provides powerful formalisms for representing and manipulating structured knowledge, but traditionally relies on explicitly defined symbols and rules and therefore faces difficulties when dealing with noisy, incomplete, or high-dimensional data. The integration of learning and reasoning has consequently become a central topic in neurosymbolic artificial intelligence [1]. Within this context, Logic Tensor Networks (LTNs) provide a framework for integrating first-order logic with differentiable neural computation. LTN introduces a many-valued, end-to-end differentiable logical language, called Real Logic, in which the elements of a first-order logic signature are grounded onto data through neural computational graphs. Logical formulas are assigned truth values in the interval $[0, 1]$, allowing logical constraints to participate directly in gradient-based learning [1]. This provides a common framework in which learning from data and reasoning with background knowledge can be expressed within the same formalism. More recently, the Structural Logic Tensor Networks (sLTN) framework extends the original LTN formulation to explicitly represent structural organization in data. While conventional LTN is primarily formulated for domains consisting of collections of individuals, many real-world datasets possess additional dimensions such as temporal order, sequence position, or graph structure. sLTN introduces

structural dimensions, structural variables, and structural relations as explicit components of the logical language, allowing such organization to be represented and reasoned about at the logical level [2]. This thesis investigates this extension from a research perspective, focusing not only on how structural information can be represented, but also on whether explicitly modelling such information can improve the generalization of neurosymbolic learning systems.

1.1 Logic Tensor Networks

Logic Tensor Networks were introduced as a neurosymbolic framework designed to combine the representational capabilities of first-order logic with the learning capabilities of neural networks. The central idea is to interpret the symbols of a logical language using differentiable tensorial representations, so that logical knowledge can be incorporated directly into a machine learning system [1]. Traditional machine learning approaches generally represent knowledge implicitly through numerical parameters learned from data. Such approaches are particularly effective when large amounts of labelled or unlabelled data are available, but the resulting representations do not necessarily expose the logical relationships that hold between concepts. Symbolic approaches, on the other hand, represent knowledge explicitly using concepts, relations, and logical rules. This makes symbolic representations suitable for reasoning and knowledge manipulation, but classical symbolic systems are generally less suited to directly processing raw, noisy, or high-dimensional observations [1]. Neurosymbolic approaches attempt to bridge this distinction by combining symbolic representations with sub-symbolic learning. In this setting, symbolic knowledge can provide constraints and relationships that guide learning, while neural models provide mechanisms for learning representations from data. A fundamental issue in such systems is therefore the grounding of symbolic entities into concrete data representations. LTN addresses this problem by assigning differentiable tensor representations and functions to the elements of a first-order logical language [1]. The LTN framework is based on Real Logic, a many-valued and differentiable extension of first-order logic. In this setting, logical formulas are not restricted to the classical truth values of true and false. Instead, their truth values are represented numerically

in the interval $[0, 1]$. Logical connectives and quantifiers are implemented using differentiable fuzzy-logic operations, allowing the satisfaction of logical formulas to be optimized using gradient-based methods [1]. This formulation leads to an important reinterpretation of learning. Rather than defining learning solely as the minimization of a task-specific numerical loss, LTN allows the learning problem to be expressed through a collection of logical formulas whose degree of satisfaction is optimized. The parameters of the underlying neural computational graphs can consequently be learned so as to maximize the overall truth value of the specified knowledge [1]. The resulting framework can express a variety of learning and reasoning tasks. The original LTN work demonstrates applications including data clustering, multi-label classification, relational learning, query answering, semi-supervised learning, regression, and embedding learning [1]. The common feature of these tasks is that data-driven learning and logical constraints can be expressed within a single differentiable framework. An LTN model can therefore be viewed as consisting of two interacting components. On the one hand, neural computation provides continuous representations and differentiable mappings from data to these representations. On the other hand, the logical layer specifies relationships and constraints between the represented entities. The grounding connects these two levels by mapping logical terms and predicates to differentiable tensor operations [1]. This combination is particularly relevant when background knowledge is available. Instead of treating the available knowledge as an external post-processing mechanism, LTN incorporates it directly into the learning objective. The logical structure can consequently influence the learned representations while the neural components provide the flexibility required to learn from numerical data [1].

1.2 From Flat Domains to Structured Data

Despite the generality of the LTN framework, its original formulation is primarily oriented towards domains that can be described as collections of individuals and relations between them. This setting is appropriate for many relational learning problems, where entities can be represented independently and logical predicates describe their properties or relationships. However, a large class of modern datasets contains an additional structural organization that

cannot be naturally described only through relations between independent individuals [2]. Examples include temporal sequences, where observations are associated with ordered time steps; sequential data, where the position of an element within a sequence is significant; and graphs, where the connectivity between nodes provides essential information about the domain. In these settings, the position or structural context of an observation is itself meaningful. Treating such information merely as an additional property of an individual does not necessarily provide a natural representation of the underlying structure [2]. Consider a temporal dataset represented as a tensor. One axis may correspond to different individuals or samples, while another axis represents time. The temporal axis has a meaning that is different from the identity of the individuals contained in the tensor. Similarly, in a sequence, the position of an element is not simply another individual in the domain: it represents its location within an ordered structure. In a graph, the organization of nodes and their connectivity introduces another form of structure that is not captured solely by treating nodes as an unordered collection [2]. This distinction motivates the introduction of explicit structural information into the logical language. The sLTN framework addresses this problem by extending LTN with structural dimensions that represent named dimensions of structured objects. Examples of structural dimensions include time steps, sequence positions, and graph nodes [2]. Structural dimensions are therefore conceptually different from ordinary logical sorts and individuals. An ordinary first-order variable ranges over individuals in a domain, whereas a structural variable ranges over positions associated with a structural dimension. This distinction allows logical formulas to refer explicitly to the organization of data rather than encoding structural information indirectly [2]. The introduction of structural dimensions also makes it possible to define structural relations. A structural relation can describe relationships between positions within a structure. For example, a relation such as

$$\text{next}(t, t')$$

can express that t' is the position immediately following t in a temporal or sequential dimension. Such relations can then be used inside logical formulas to express constraints involving

neighbouring or otherwise structurally related positions [2]. This extension is important because structural relationships can influence the interpretation of a formula. A constraint may not simply state that a property holds for an individual, but that it holds at a particular structural position, or that properties at two positions are related according to the structure of the underlying data. sLTN consequently provides a way to express temporal, sequential, and relational constraints directly within the logical language [2].

1.3 Structural Logic Tensor Networks

Structural Logic Tensor Networks extend the LTN framework by making structural dimensions first-class elements of the logical language. The framework introduces an explicit distinction between the logical domain of individuals and the structural organization associated with tensors. This allows logical expressions to refer to positions within structured data while retaining the differentiable fuzzy semantics of LTN [2]. The formal language of sLTN includes ordinary first-order elements such as sorts, constants, variables, functions, and predicates, together with structural elements such as structural variables and structural relations. Structural dimensions provide the link between the logical description and the axes of the underlying tensors [2]. A structural dimension can be understood as a named axis associated with a particular type of organization in the data. For example, a dimension called *time* may identify the temporal axis of a tensor, while another dimension may represent positions in a sequence. The structural dimension is not merely metadata: it participates in the interpretation of logical expressions by determining how tensorial data are aligned, selected, masked, or aggregated [2]. Structural variables are used to quantify over the positions of these dimensions. In contrast to ordinary first-order variables, which range over individuals, structural variables range over indices associated with structural dimensions. This distinction allows formulas to make explicit statements about the location of data within a structured object [2]. Structural relations provide a further level of expressiveness. They allow relationships between structural positions to be represented explicitly. For example, a temporal relation can identify consecutive time steps, while other structural relations can describe more general dependencies between positions. These relations can be Boolean or

fuzzy and can therefore be integrated into the differentiable semantics of the framework [2]. An important consequence of this design is that structural information becomes part of the logical theory rather than being handled exclusively by the underlying neural architecture. The logical language can therefore express constraints whose meaning depends directly on the organization of the data. This is particularly relevant for problems in which temporal or sequential dependencies are part of the knowledge that should guide learning [2]. The sLTN framework retains the central principle of LTN: logical expressions are interpreted through differentiable tensor operations. Thus, introducing structural dimensions does not replace the original tensorial semantics but extends it to account for additional axes of organization. The resulting framework can align logical expressions with tensor dimensions and use structural information when performing selection, masking, or aggregation operations [2]. An important theoretical property of sLTN is that the original LTN framework can be recovered as a special case. When no structural dimensions are introduced, the structural extension does not add additional structure to the interpretation, and the framework reduces to the original LTN setting [2].

1.4 Syntax, Semantics, and Tensorial Interpretation

A central aspect of sLTN is the explicit separation between the syntactic and semantic levels of the framework. The language first defines the symbols and expressions that are permitted by the theory. These expressions are then parsed and validated before being interpreted over tensorial data [2]. The signature specifies the vocabulary available to a theory. It contains the sorts, dimensions, constants, variables, functions, predicates, and structural relations required to construct formulas. This explicit signature provides a formal description of the entities and relationships that can appear in a logical theory [2]. The syntactic layer is followed by formula parsing and validation. This separation is important because it distinguishes the logical structure of an expression from the numerical representation used to evaluate it. A formula can therefore be treated as a symbolic object before being mapped to tensor operations by the interpretation mechanism [2]. At the semantic level, the symbols of the logical language are assigned tensorial interpretations. Constants and variables are

grounded into tensor representations, while predicates and functions are interpreted through differentiable operations. Logical connectives and quantifiers are then evaluated using fuzzy semantics, producing numerical truth values [1, 2]. Structural dimensions introduce an additional component into this interpretation. A structural dimension determines an axis of the tensorial representation, while structural variables identify positions along that axis. Structural relations determine relationships between such positions. The interpretation can consequently use the structural information to align tensors and to perform operations that respect the organization encoded by the data [2]. This tensorial interpretation is essential for maintaining differentiability. Logical reasoning is not implemented as a separate symbolic process disconnected from learning. Instead, formulas are translated into differentiable numerical operations whose outputs can contribute to an optimization objective. As in LTN, the truth value of a formula can therefore influence the learning of the parameters involved in its grounding [1, 2]. The separation between syntax and semantics also provides a modular view of the framework. The logical theory specifies what can be expressed, while the interpretation specifies how those expressions are connected to the underlying data. This distinction becomes particularly useful for structured data because the same logical concepts can be associated with different tensorial organizations through their structural dimensions [2].

1.5 Motivation and Research Gap

The introduction of structural dimensions and relations provides a mechanism for representing information that is intrinsic to the organization of structured data. However, the availability of such a representation does not by itself establish whether explicit structural reasoning provides a measurable advantage during learning. In particular, it remains important to determine the conditions under which structural information can improve the ability of a neurosymbolic model to generalize learned knowledge to structural configurations that were not observed during training. This question is particularly relevant because a model may achieve good predictive performance without explicitly representing structure. A purely neural model may learn statistical regularities from the observations, while a conventional

LTN may incorporate logical constraints without explicitly modelling structural positions and relations. Consequently, any improvement observed with sLTN should not automatically be attributed to the presence of explicit structural reasoning. This thesis therefore focuses on the relationship between structural information and generalization. Rather than assuming that explicit structural reasoning is beneficial in every setting, the study investigates whether its contribution depends on the extent to which the target task relies on structural relationships. In addition, the study examines whether this relationship changes when the amount of available training data is reduced.

The central research problem can therefore be formulated as follows:

When does explicit structural reasoning improve the generalization of neurosymbolic models to previously unseen structural configurations?

The experimental analysis is designed to identify conditions under which explicit structural representations provide a measurable benefit, as well as conditions in which their contribution is limited. This allows the study to characterize the role of structural reasoning rather than simply comparing the overall predictive performance of different models.

1.6 Research Questions

The main objective of this thesis is to investigate the conditions under which explicitly representing structural information can improve the ability of neurosymbolic models to generalize beyond the structural configurations observed during training. To address this objective, the thesis considers the following research questions:

- **RQ1 — Structural Generalization.**

To what extent does explicit structural reasoning affect the generalization of neurosymbolic models to previously unseen structural configurations?

- **RQ2 — LTN and sLTN Comparison.**

How does the generalization performance of sLTN compare with that of conventional LTN when the structural configurations encountered during testing differ from those observed during training?

- **RQ3 — Conditions for Structural Benefit.**

How does the benefit of explicit structural reasoning vary with the structural dependency of the task and the amount of available training data?

These research questions are addressed through a controlled experimental evaluation involving a neural baseline, a conventional Logic Tensor Network, and a Structural Logic Tensor Network incorporating explicit structural dimensions and relations.

1.7 Research Hypotheses

Based on the research questions introduced in the previous section, this thesis formulates the following hypotheses to be empirically investigated:

- **H1 — Structural Generalization.**

Explicit structural representations are expected to improve generalization to previously unseen structural configurations when the target task depends on structural relationships.

- **H2 — Structural Dependency.**

The relative benefit of explicit structural reasoning is expected to increase with the degree of structural dependency of the task. Conversely, when the target task can be solved primarily from local or non-structural information, the difference between models with and without explicit structural representations is expected to be smaller.

- **H3 — Data Efficiency.**

The benefit of explicit structural reasoning is expected to become more pronounced when the amount of available training data is reduced, particularly for tasks that strongly depend on structural relationships.

These hypotheses are formulated as empirical research hypotheses rather than assumptions about the superiority of sLTN. The experimental evaluation therefore considers both scenarios in which explicit structural reasoning provides a measurable benefit and scenarios in which its contribution is limited or absent. The objective is to identify the conditions associated with a structural advantage rather than to establish a priori that one model is universally superior to another.

1.8 Contributions of the Thesis

The primary contribution of this thesis is a systematic experimental investigation of the conditions under which explicit structural representations contribute to structural generalization in neurosymbolic learning. The experimental framework provides a controlled comparison between three model settings: a neural baseline, a conventional Logic Tensor Network, and a Structural Logic Tensor Network incorporating explicit structural dimensions and relations. This comparison provides a basis for separating the contribution of logical constraints from that of explicit structural representations. A further contribution is the design of controlled structural generalization scenarios in which the structural configurations encountered during evaluation differ from those observed during training. This experimental setting provides a direct means of assessing whether models can generalize learned knowledge across changes in structural organization. The thesis also investigates how the contribution of explicit structural reasoning changes with the structural dependency of the task. By considering tasks with different degrees of dependence on structural relationships, the study aims to characterize the conditions under which explicit structural representations become useful and the conditions under which their contribution is limited. Finally, the experimental evaluation considers different amounts of available training data. This analysis investigates whether the effect of explicit structural reasoning changes as supervision becomes more limited, with particular attention to tasks in which structural relationships are important for the target prediction.

1.9 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2** introduces the theoretical background required for the study, including neurosymbolic learning, first-order logic, fuzzy logic, Logic Tensor Networks, and the principles underlying structured data representations.
- **Chapter 3** presents the Structural Logic Tensor Network framework in greater detail, focusing on its syntax, semantics, structural dimensions, structural variables, structural relations, and tensorial interpretation.
- **Chapter 4** describes the research methodology adopted in this thesis. It introduces the experimental task and dataset, the considered models, the structural configurations used during training and testing, the different levels of structural dependency, the amounts of training data considered, the evaluation metrics, and the experimental protocols used to address the research questions.
- **Chapter 5** presents the experimental evaluation of structural generalization. The chapter investigates the performance of the considered models on structural configurations observed during training as well as on previously unseen configurations. It also examines model performance under different levels of structural dependency and different amounts of training data.
- **Chapter 6** analyses the experimental results, with particular emphasis on the relationship between structural dependency, generalization performance, logical satisfaction, and the amount of training data required to achieve comparable performance.
- **Chapter 7** discusses the findings in relation to the research questions and hypotheses. Particular attention is given to identifying the conditions under which explicit structural reasoning provides measurable benefits, as well as the conditions under which its contribution is limited.

- **Chapter 8** summarizes the main findings of the thesis, discusses its limitations, and outlines directions for future research, including the potential extension of the experimental analysis to additional structured domains and neurosymbolic learning settings.

Chapter 2

Theoretical Background

Chapter 3

Structural Logic Tensor Networks

sLTN in detail: structural dimensions, structural variables/relations, syntax, semantics,
tensor interpretation

Chapter 4

Experimental Methodology

Actual research design: task, data, models, train/test structures, structural dependency, data amounts, metrics

Chapter 5

Experimental Results

What happened in the experiments — tables, plots, accuracy/F1, generalization results

Chapter 6

Analysis of Structural Generalization

Interpret the results specifically in terms of when structural reasoning helps

Chapter 7

Discussion

Answer RQs/Hypotheses, compare findings with LTN/sLTN literature, explain implications and limitations

Chapter 8

Conclusions and future research

Final findings, contribution, limitations, future research

Bibliography

Articles and papers

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